STUDENT REFERENCE & INTERNAL USE NOTICE

This protocol documents standardized in-vitro Antimicrobial Susceptibility Testing (AST) and antimicrobial activity screening procedures (agar disc diffusion, agar well diffusion, and broth microdilution MIC) used in HOPE Lab research and is PROVIDED ONLY FOR THE HOPE LAB MEMBERS for training and internal reference. External users should independently assess method suitability, verify current standards and chemical safety for reagents, and obtain required biosafety authorization and other essential approvals. This document does not replace an institutionally approved SOP or method-specific risk assessment.

TITLE

Antimicrobial Susceptibility Testing (AST) and In-Vitro Antimicrobial Activity Assays

PRINCIPLE

Antimicrobial Susceptibility Testing (AST) evaluates the in-vitro growth-inhibitory response of a microorganism exposed to defined concentrations of antimicrobial agents, whereas antimicrobial activity screening evaluates whether candidate probiotic supernatants, bacteriocins, or natural extracts inhibit indicator pathogens. Two complementary quantitative and semi-quantitative methodologies are utilized:

• Agar Diffusion (Kirby–Bauer Disc Diffusion & Agar Well Diffusion): An antimicrobial agent diffuses radially from a calibrated paper disc or agar well through a standardized agar matrix seeded with a uniform bacterial lawn (0.5 McFarland). Where the local concentration exceeds the organism's inhibitory threshold, a circular zone of growth inhibition forms.

• Broth Microdilution (MIC Determination): Two-fold serial dilutions of an antimicrobial agent in standardized liquid medium are inoculated with a defined bacterial density (~5×10^5 CFU/mL). The Minimum Inhibitory Concentration (MIC) is the lowest concentration that completely inhibits visible microbial growth after standardized incubation.

Zone diameters and MIC values depend critically on medium composition, agar depth (4.0 ± 0.5 mm), inoculum density, incubation atmosphere/duration, and agent molecular weight/diffusivity. Agar diffusion zone diameters cannot be compared directly across structurally different compounds with distinct diffusion rates, and in-vitro susceptibility does not automatically guarantee clinical therapeutic efficacy in vivo.

Learning Outcomes

- Distinguish standardized clinical antibiotic susceptibility testing (disc diffusion & broth microdilution MIC) from agar well diffusion screening of biological extracts/supernatants.
- Standardize bacterial inocula accurately to 0.5 McFarland (~$1–2 \times 10^8$ CFU/mL) using spectrophotometric or turbidity standards.
- Control critical physical variables including Mueller–Hinton agar depth, cation supplementation, disc spacing, and pre-diffusion timing.
- Measure inhibition zone diameters accurately in millimeters and determine MIC endpoints against sterility and growth controls.
- Apply CLSI/EUCAST quality-control reference strains and avoid invalid cross-compound zone-size comparisons.

OBJECTIVE & SCOPE

Objective: To establish standardized procedures, inoculum controls, endpoint reading criteria, and interpretive limitations for disc diffusion, agar well diffusion, and broth microdilution MIC assays.

Scope: Applicable to non-fastidious aerobic/facultative bacteria tested on Mueller–Hinton media and lactic acid bacteria tested on validated ISO 10932 / LSM (LAB Susceptibility Test Medium) formulations. Fastidious pathogens (e.g., Streptococcus pneumoniae, Neisseria, anaerobes) require specific supplemented media and CO_2 conditions per current CLSI/EUCAST standards.

CORE CONCEPTS & METHOD SELECTION

- **0.5 McFarland Inoculum Standardization:** A 0.5 McFarland turbidity standard corresponds to an optical density at 625 nm (OD_{625}) of 0.08–0.13 (1 cm path length), representing approximately 1–2 × 10^8 CFU/mL for E. coli ATCC 25922. Over-inoculation artificially shrinks inhibition zones and elevates MICs; under-inoculation causes false susceptibility.
- **Disc Diffusion vs. Agar Well Diffusion:** Standardized commercial antibiotic discs (6 mm) follow calibrated CLSI/EUCAST breakpoints on Mueller–Hinton Agar (MHA). Agar well diffusion (typically 6–8 mm wells loaded with 50–100 µL liquid) is used for screening cell-free supernatants, neutralized organic acids, or extracts where no commercial disc exists; results are reported in millimeters (mm) rather than clinical S/I/R categories.
- **MIC & MBC Definitions:** Minimum Inhibitory Concentration (MIC) is the lowest concentration (µg/mL or mg/mL) preventing visible turbidity in broth microdilution. Minimum Bactericidal Concentration (MBC) is determined by subculturing clear MIC wells onto drug-free agar and identifying the lowest concentration yielding ≥99.9% (3-log_{10}) kill relative to the initial inoculum.
- **Mandatory Solvent & Acid Controls:** When testing extracts dissolved in DMSO/ethanol, include a solvent-only vehicle control at the highest tested concentration (keep DMSO ≤1% v/v in broth). When screening LAB supernatants for bacteriocin activity, test both native and pH-neutralized (pH 6.5–7.0) supernatants to distinguish organic-acid inhibition from proteinaceous antimicrobials.

MATERIALS & EQUIPMENT

Reagents & Consumables

- Mueller–Hinton Agar (MHA) & Cation-Adjusted Mueller–Hinton Broth (CAMHB) (or ISO-10932 LSM for LAB)
- Calibrated commercial antimicrobial discs (6 mm) & analytical-grade antibiotic powders
- 0.5 McFarland turbidity standard (barium sulfate or latex suspension) & sterile 0.85% saline
- Reference QC strains (e.g., E. coli ATCC 25922, S. aureus ATCC 25923/29213, P. aeruginosa ATCC 27853)
- Uniform 90 mm Petri dishes (poured to 4.0 ± 0.5 mm depth), sterile 96-well U/flat-bottom microplates, & sterile cork borer

Equipment & Instrumentation

- Class II Biosafety Cabinet (BSL-2)
- Calibrated Spectrophotometer (OD_{625} nm) or Densitometer
- Bacteriological Incubator calibrated to 35 ± 2°C
- Digital Vernier Calipers (0.1 mm resolution) & non-reflective dark viewing surface
- Multichannel Micropipettes, Sterile Forceps, & Microplate Reader (optional turbidity check)

STEP-BY-STEP PROCEDURE

Step 1: Inoculum Preparation & Standardization (Direct Colony Suspension)

Select 3–5 morphologically identical colonies from an 18–24 h non-selective agar plate and suspend in sterile 0.85% saline. Vortex thoroughly and adjust turbidity to match a 0.5 McFarland standard (OD_{625} = 0.08–0.13, ~1–2 × 10^8 CFU/mL). Use the standardized suspension within 15 minutes of preparation.

Step 2: Agar Lawn Inoculation (Disc & Well Diffusion)

Dip a sterile cotton swab into the 0.5 McFarland suspension, rotate firmly against the upper inside tube wall to remove excess fluid, and streak evenly across the entire surface of a pre-dried MHA plate (poured to 4.0 ± 0.5 mm depth) in three directions (rotating 60° each time), finishing with a sweep around the agar rim. Allow plates to dry with lids ajar for 3–5 minutes (no more than 15 min).

Step 3A: Kirby–Bauer Disc Placement (For Calibrated Antibiotics)

Using sterile forceps or a disc dispenser, place antibiotic discs firmly onto the inoculated agar surface so complete contact is achieved. Space discs ≥24 mm center-to-center (maximum 5–6 discs per 90 mm plate) and ≥15 mm from the plate edge. Never relocate a disc once it touches the agar surface.

Step 3B: Agar Well Diffusion Setup (For Supernatants / Extracts)

Punch uniform circular wells (6–8 mm diameter) into the seeded agar using a sterile cork borer and remove agar plugs aseptically. Dispense a fixed volume (e.g., 50–100 µL) of filter-sterilized test supernatant (native and pH-neutralized) or extract into each well, alongside a matched vehicle/medium negative control well and a positive antibiotic control well. Allow 30–60 minutes at 4°C or room temperature for radial pre-diffusion before inverting.

Step 4: Broth Microdilution MIC Setup (96-Well Format)

Dispense 50 µL of CAMHB into columns 2–12 of a sterile 96-well plate. Add 100 µL of 2× working antimicrobial stock into column 1, then perform two-fold serial dilutions across columns 1–10 (discarding 50 µL from column 10). Reserve column 11 as the Positive Growth Control (broth + inoculum, no drug) and column 12 as the Sterility Control (broth only). Dilute the 0.5 McFarland suspension 1:100 in CAMHB (~1 × 10^6 CFU/mL) and add 50 µL to columns 1–11 (final inoculum ~5 × 10^5 CFU/mL).

Step 5: Incubation & Endpoint Measurement

Incubate agar plates (inverted) and microplates (covered/sealed to prevent evaporation) at 35 ± 2°C for 16–20 hours (24–48 h for LAB). Measure complete inhibition zone diameters (including the 6 mm disc/well) in millimeters with digital calipers against a dark, non-reflective background. For MIC plates, verify clear Sterility Control and turbid Growth Control, then record the lowest concentration with no visible growth button/turbidity.

CRITICAL FACTORS AFFECTING RESULTS

Variable	Effect of Inadequate Control	Required Control Standard
Inoculum Density	Heavy inoculum shrinks zones/raises MIC; light inoculum enlarges zones	Standardize to 0.5 McFarland (OD_{625} = 0.08–0.13); use within 15 min
Agar Depth & Composition	Thick agar dilutes drug diffusion; excess thymidine/cations antagonize drugs	Pour MHA to exact 4.0 ± 0.5 mm depth; use validated CAMHB/MHA lots
Disc-to-Agar Contact	Poor contact delays diffusion and produces smaller/irregular zones	Press discs gently onto agar immediately after placement; never move
Supernatant Acidity (LAB)	Lactic/acetic acid creates halos mistaken for bacteriocin activity	Test parallel pH-neutralized (pH 6.5–7.0) and catalase-treated aliquots

Vehicle Solvent Toxicity	High DMSO/ethanol kills indicator bacteria independently of extract	Include vehicle-only control; keep DMSO ≤1% v/v in broth assays
Reading & Breakpoints	Reading partial microcolonies or outdated breakpoints misclassifies strains	Measure complete inhibition; cite exact CLSI/EUCAST edition & year

QUALITY CONTROL & TROUBLESHOOTING

Quality Control (QC): Test standard ATCC reference strains (e.g., E. coli ATCC 25922, S. aureus ATCC 25923/29213) alongside every AST run. Zone diameters and MIC values for QC strains must fall within published CLSI/EUCAST QC ranges before test-strain data are accepted.

Observation	Possible Explanation	Corrective Action
Patchy or non-confluent lawn	Inoculum too dilute, swab too dry, or uneven streaking	Re-verify 0.5 McFarland OD; streak in 3 rotated directions
Overlapping inhibition zones	Discs/wells placed too close together or overly potent stock	Space discs/wells ≥24 mm center-to-center; dilute extract
Double zone / inner haze	Heteroresistant subpopulation, swarming (Proteus), or sulfonamide haze	Measure inner complete-inhibition ring; ignore ≤20% sulfonamide haze
Growth in MIC sterility well	Aseptic contamination during multichannel pipetting or broth prep	Invalidate plate; repeat with fresh sterile CAMHB and tips
QC strain out of range	Degraded antibiotic disc/stock, wrong medium pH, or wrong temp	Replace antibiotic stock/discs; verify incubator 35°C & MHA pH 7.2–7.4

BIOSAFETY & ADDITIONAL CONSIDERATIONS

- **Biosafety & Containment:** Handle all indicator pathogens (e.g., S. aureus, Salmonella, P. aeruginosa, E. coli) under strict BSL-2 containment inside a Class II Biosafety Cabinet. Autoclave all inoculated AST plates, microplates, saline tubes, and swabs at 121°C for 30 minutes prior to disposal.
- **Scientific Interpretation Boundary:** Do not equate larger agar diffusion zones across different antimicrobial compounds with higher potency—molecular weight, solubility, and agar charge interactions govern diffusion rate. Always confirm comparative potency using broth microdilution MIC values.

REFERENCES & RELATED DOCUMENTS

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